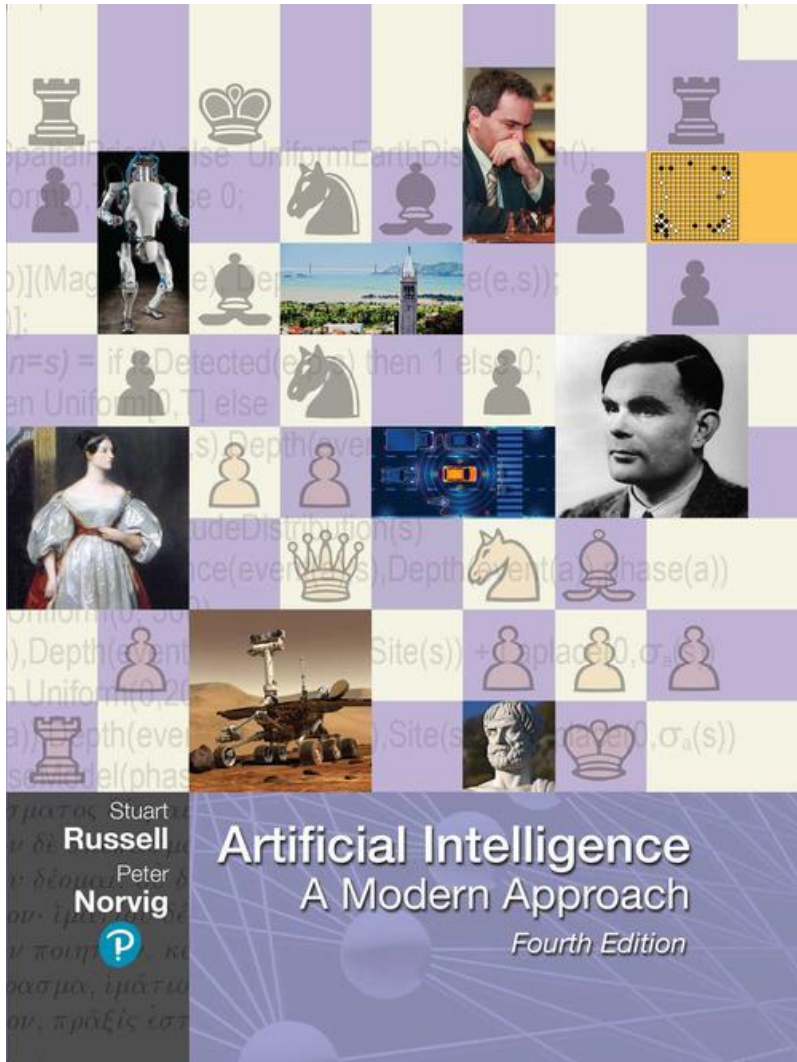


Artificial Intelligence: A Modern Approach

Fourth Edition



Chapter 1

Introduction

Artificial Intelligence

Chapter 1

Outline

- ◆ What is AI?
- ◆ A brief history
- ◆ The state of the art

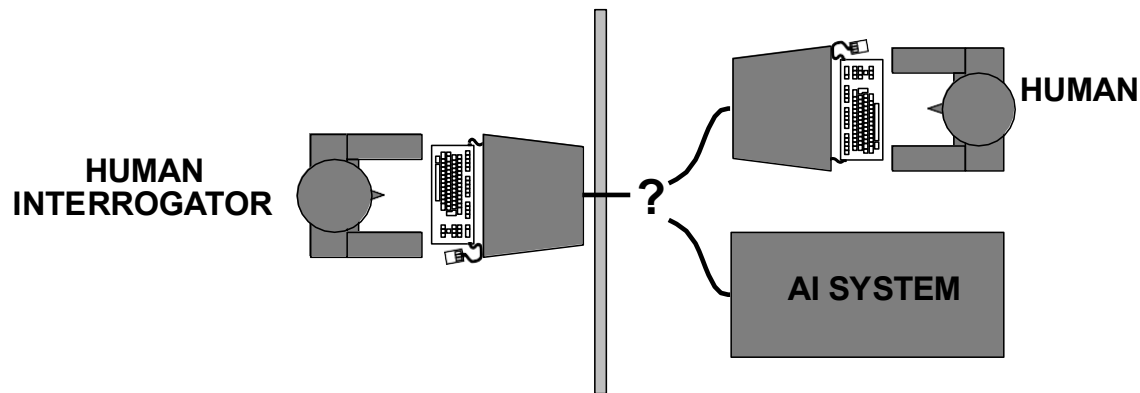
What is AI?

Systems that think like humans	Systems that think rationally
Systems that act like humans	Systems that act rationally

Acting humanly: The Turing test

Turing (1950) “Computing machinery and intelligence”:

- ◆ “Can machines think?” → “Can machines behave intelligently?”
- ◆ Operational test for intelligent behavior: the **Imitation Game**



- ◆ Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- ◆ Anticipated all major arguments against AI in following 50 years
- ◆ Suggested major components of AI: knowledge, reasoning, language understanding, learning

Problem: Turing test is not **reproducible**, **constructive**, or amenable to **mathematical analysis**

Thinking humanly: Cognitive Science

1960s “cognitive revolution”: information-processing psychology replaced prevailing orthodoxy of behaviorism

Requires scientific theories of internal activities of the brain

- What level of abstraction? “Knowledge” or “circuits”?
- How to validate? Requires
 - 1) Predicting and testing behavior of human subjects (top-down)
 - or 2) Direct identification from neurological data (bottom-up)

Both approaches (roughly, Cognitive Science and Cognitive Neuroscience) are now distinct from AI

Both share with AI the following characteristic:

the available theories do not explain (or engender)
anything resembling human-level general intelligence

Hence, all three fields share one principal direction!

Thinking rationally: Laws of Thought

Normative (or prescriptive) rather than descriptive

Aristotle: what are correct arguments/thought processes?

Several Greek schools developed various forms of logic:

notation and rules of derivation for thoughts;
may or may not have proceeded to the idea of mechanization

Direct line through mathematics and philosophy to modern AI

Problems:

- 1) Not all intelligent behavior is mediated by logical deliberation
- 2) What is the purpose of thinking? What thoughts should I have out of all the thoughts (logical or otherwise) that I could have?

Acting rationally

Rational behavior: doing the right thing

The right thing: that which is expected to maximize goal achievement, given the available information

Doesn't necessarily involve thinking—e.g., blinking reflex—but thinking should be in the service of rational action

Aristotle (Nicomachean Ethics):

Every art and every inquiry, and similarly every action and pursuit, is thought to aim at some good

Rational agents

An **agent** is an entity that perceives and acts

This course is about designing **rational agents**

Abstractly, an agent is a function from percept histories to actions:

$$f : P^* \rightarrow A$$

For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance

Caveat: **computational limitations make perfect rationality unachievable**

→ design best **program** for given machine resources

AI prehistory

Philosophy	logic, methods of reasoning mind as physical system foundations of learning, language, rationality
Mathematics	formal representation and proof algorithms, computation, (un)decidability, (in)tractability probability
Psychology	adaptation phenomena of perception and motor control experimental techniques (psychophysics, etc.)
Economics	formal theory of rational decisions
Linguistics	knowledge representation grammar
Neuroscience	plastic physical substrate for mental activity
Control theory	homeostatic systems, stability simple optimal agent designs

Potted history of AI

- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1952–69 Look, Ma, no hands!
- 1950s Early AI programs, including Samuel's checkers program,
Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1965 Robinson's complete algorithm for logical reasoning
- 1966–74 AI discovers computational complexity
Neural network research almost disappears
- 1969–79 Early development of knowledge-based systems
- 1980–88 Expert systems industry booms
- 1988–93 Expert systems industry busts: "AI Winter"
- 1985–95 Neural networks return to popularity
- 1988– Resurgence of probability; general increase in technical depth
"Nouvelle AI": ALife, GAs, soft computing
- 1995– Agents, agents, everywhere . . .
- 2003– Human-level AI back on the agenda

State of the art

Which of the following can be done at present?

- ♦ Play a decent game of table tennis

State of the art

Which of the following can be done at present?

- ◆ Play a decent game of table tennis
- ◆ Drive safely along a curving mountain road

State of the art

Which of the following can be done at present?

- ◆ Play a decent game of table tennis
- ◆ Drive safely along a curving mountain road
- ◆ Drive safely along Telegraph Avenue

State of the art

Which of the following can be done at present?

- ◆ Play a decent game of table tennis
- ◆ Drive safely along a curving mountain road
- ◆ Drive safely along Telegraph Avenue
- ◆ Buy a week's worth of groceries on the web

State of the art

Which of the following can be done at present?

- ◆ Play a decent game of table tennis
- ◆ Drive safely along a curving mountain road
- ◆ Drive safely along Telegraph Avenue
- ◆ Buy a week's worth of groceries on the web
- ◆ Buy a week's worth of groceries at Berkeley Bowl

State of the art

Which of the following can be done at present?

- ◆ Play a decent game of table tennis
- ◆ Drive safely along a curving mountain road
- ◆ Drive safely along Telegraph Avenue
- ◆ Buy a week's worth of groceries on the web
- ◆ Buy a week's worth of groceries at Berkeley Bowl
- ◆ Play a decent game of bridge

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Which of the following can be done at present?

- ◆ Play a decent game of table tennis
- ◆ Drive safely along a curving mountain road
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- ◆ Buy a week's worth of groceries on the web
- ◆ Buy a week's worth of groceries at Berkeley Bowl
- ◆ Play a decent game of bridge
- ◆ Discover and prove a new mathematical theorem

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- ◆ Discover and prove a new mathematical theorem
- ◆ Design and execute a research program in molecular biology

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- ◆ Write an intentionally funny story

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- ◆ Design and execute a research program in molecular biology
- ◆ Write an intentionally funny story
- ◆ Give competent legal advice in a specialized area of law

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- ◆ Discover and prove a new mathematical theorem
- ◆ Design and execute a research program in molecular biology
- ◆ Write an intentionally funny story
- ◆ Give competent legal advice in a specialized area of law
- ◆ Translate spoken English into spoken Swedish in real time

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- ◆ Write an intentionally funny story
- ◆ Give competent legal advice in a specialized area of law
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- ◆ Converse successfully with another person for an hour

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- ◆ Converse successfully with another person for an hour
- ◆ Perform a complex surgical operation

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- ◆ Design and execute a research program in molecular biology
- ◆ Write an intentionally funny story
- ◆ Give competent legal advice in a specialized area of law
- ◆ Translate spoken English into spoken Swedish in real time
- ◆ Converse successfully with another person for an hour
- ◆ Perform a complex surgical operation
- ◆ Unload any dishwasher and put everything away

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Risks and Benefits of AI

“First solve AI, then use AI to solve everything else.” Demis Hassabis, CEO of Google DeepMind

Benefits:

- Decrease repetitive work
- Increase production of goods and services
- Accelerate scientific research (disease cures, climate change and resource shortages solutions)

Risks:

- Lethal autonomous weapons
- Surveillance and persuasion
- Biased decision making
- Impact on employment
- Safety-critical applications
- Cybersecurity threats

Risks and Benefits of AI

Development of an artificial superintelligence that surpasses human intelligence may pose a significant risk

Analogous to the “Gorilla problem”

Humans and gorillas evolved from the same species, but humans have more control than other primates.

Thus, we should design AI systems in such a way that they do not end up taking control in the way that Turing suggests they might.